

Notation Guide

Jason Karpeles | Constructed exercises and conditional calculations

Chapter links in this companion copy open the corresponding laboratory reading page. The symbols and explanations are retained from the book; its original notation source is preserved in [reference-sources/notation-guide.md](#).

Core symbols carry their meanings across chapters. Explicitly declared local aliases and diagram labels are scoped where they appear; a local count or vertex label does not redefine a global quantity.

That rule costs something. It means a few symbols are less conventional than the ones a specialist in one subfield would expect, because a letter that is standard in information theory may already be doing a job in decision theory. Where this book departs from a field's usual notation, it is to avoid a collision, and the departure is noted at the point of use.

This guide is for returning to, not for memorizing. Every symbol is also defined where it first appears.

Conventions

- Scalars are italic; vectors are bold lowercase; matrices are uppercase.
- Sets, spaces, and operators use calligraphic or roman forms.
- Random variables are uppercase and their realized values are the matching lowercase letter, so the composite state random variable is written one way and a particular value of it another.
- Named operations such as `dec`, `Mem`, and `Reach` are roman, never italic single letters.
- Probability is $\Pr$ and expectation is $\mathbb{E}$. Chapters 1 and 22 typeset the expectation operator as $\mathbb{E}$; Chapters 6, 7, 10, 11, 13, 15, 16, and 19 typeset the same operator as roman E (E or $\mathcal{E}$). Both forms denote one operator. A plain E used as a subscript label, such as a delegated destination in Chapter 27, names a thing and is not an operator.
- Every logarithm states its base or its units at first use.
- Subscript t is time, except where a chapter declares otherwise (Chapter 19's matrix $M_{\{st\}}$ uses s and t as run indices). Subscript m indexes a member of a declared model family.
- A few letters change meaning between chapters, for example Γ , α , δ , τ , C , z , u , Φ , and λ . Each meaning is scoped to the chapter listed beside it and restated where it applies.

The two quantities the book is built on

Two objects recur from Part I onward and are worth reading before Chapter 1.

The score of an assembly. $U_{\mathcal{V}}(S) = \mathbb{E}[Y_S \mid \mathcal{V}]$ is the population score of the system built from enabled component set S , under declared evaluation contract $\mathcal{V}$; Y_S is the score of one run of that system. $\hat{U}_{\mathcal{V}}(S)$ is its finite-sample estimate. The contract fixes task distribution, scoring rule and units, sampling settings, resource budget, and stopping rule. When the set holds only the frozen model, U describes the model-alone system. As

components are added to the set, the same notation carries through unchanged, which is the point of writing it this way.

The interaction contrast. $\Gamma(A; B)$ is the signed difference between the joint gain and the sum of isolated gains under one matched evaluation contract. Four scores identify this nonadditivity, not a mechanism or the necessity of either component. Chapter 2 defines it. ρ_Γ divides the contrast by the positive total joint gain. Nonnegative isolated gains and a nonnegative contrast additionally make the ratio a fraction between zero and one. Negative contrasts remain meaningful signed quantities.

Part I symbols

Symbol	Meaning	Introduced
$K_{\theta}(z \mid c)$	Model conditional law over outputs given context	1
$U_{\mathcal{V}}(S)$	Population task score of assembly S	1
$\hat{U}_{\mathcal{V}}(S)$	Finite-sample estimate of that task score	1
dec	Decoding map from a distribution to a candidate output	1
eval_{τ}	Evaluator with threshold, parser, or tolerance τ	1
θ	Model parameters, frozen unless a chapter says otherwise	1
$\mathbb{E}$	Expectation, as in Chapters 1 and 22 (Chapters 6, 7, 10, 11, 13, 15, 16, and 19 write the same operator as roman E; a plain subscript label such as Chapter 27's p_E is not this operator)	1
$\mathcal{V}$	Declared evaluation contract: task distribution, scoring rule and units, sampling settings, budget, and stopping rule	1
q, n	Per-token success probability and target length in the exact-match example, so the score is q^n	1
$\Gamma(A; B)$	Interaction contrast between component sets	2
ρ_{Γ}	Interaction share of total joint gain	2
q_m	Per-attempt success probability for family member m	3
$\text{pass}_{\tau}(q_m)$	Thresholded pass indicator on q_m	3
$L(C) = aC^b + c$	Fitted pretraining-loss curve; C is training compute and a, b, c are fitted constants	3
ψ	Fitted forecast parameters	3
f_{ψ}, ϵ_m	Fitted forecast function and its error for member m	3
Γ_m	Interaction contrast measured for family member m	3
z_m	Declared predictor features for member m	3
G	Action graph: vertices are controller states, edges are allowed transitions	4
V_0	Initial vertex set	4
step	One-transition successor map on vertex sets	4
$\text{Reach}_G(V_0)$	Reachable set	4

Symbol	Meaning	Introduced
$\text{Act}(v)$	Candidate actions at vertex v	4
$\mathcal{G}$	Permission grant rule: maps a set of candidate actions to the permitted subset	4
$\text{Allowed}_{\mathcal{G}}(v)$	Permitted actions after filtering	4
$\Pi_{G,v}(p)$	Infinite-cluster probability	4
$p_c(G, v)$	Critical probability	4
d	Branching factor or degree	4
$x_t = (c_t, m_t, w_t, b_t)$	Composite agent state realization	5
w_t	World or environment state	5
$P(x_{t+1} \mid x_t)$	Composite transition kernel	5
Mem	Memory map from the whole history to the retained memory m_t ; the update itself is part of Upd	5
$\pi_{\theta}(z, a \mid x)$	Proposal policy: model law times decoder	5
κ	Coarse-graining map	5
$H(X), I(X; Y)$	Shannon entropy, mutual information	5
$\text{Grant}(g \mid x, u)$	Normalized permission-outcome law, including denial	5
$\text{Env}(w', o \mid x, u, g)$	Joint next-world and observation law	5
$\text{Upd}(c', m', b' \mid x, u, g, w', o)$	Joint context, memory, and budget update law	5
$P_{\text{env}}(x' \mid x, u)$	Controlled transition given a full proposal	5
$P_{\theta}(x' \mid x)$	Closed-loop transition after averaging over proposal policy	5
$u = (z, a)$	Full proposal: model output and decoded action	5

Symbols introduced after Part I

Symbol	Meaning	Introduced
$u(y)$	Utility of a single outcome	6
y	An outcome of an action	6
$\mathcal{A}$	The action set available at a state	6
$EU(a)$	Expected utility of an action	6
CE	Certainty equivalent	6
Ret_t	Return, the discounted sum of future rewards	7
r_t	Reward received on the transition out of x_t	7
γ	Discount factor	7
$\pi(a \mid x)$	Stochastic policy	7
V^π, V^*	State value under a policy, and under an optimal policy	7
Q^π	Action value under a policy	7
$\Phi(x)$	Local shaping potential	7
$\mathcal{X}$	The finite state set	8
$\mathbf{b}_t$	Belief, a distribution over $\mathcal{X}$ at time t	8
$\mathcal{B}$	The belief simplex, the set of distributions over $\mathcal{X}$	8
$Obs(o \mid x, a)$	Observation kernel: the chance of seeing o on arriving in x after a	8
$\bar{r}(\mathbf{b}, a)$	Expected reward of action a under belief $\mathbf{b}$	8
$J(x', o \mid x, a)$	Filtered joint next-state and observation law	8
Γ_H, α	Finite-horizon policy-vector set and one vector	8
$VOI(o)$	Value of observing o before a declared decision; Chapter 27 writes $VOI(o)$ for one realized observation	8
$g(v), h(v)$	True cost from the start to vertex v , and true cost from v to the nearest goal	9
$f(v)$	Cost of the best path constrained to pass through v	9
$\hat{g}(v), \hat{h}(v), \hat{f}(v)$	The computable estimates of each	9

Symbol	Meaning	Introduced
$\text{cost}(u, v)$	Cost of the edge from u to v	9
$s, \mathcal{T}$	Start vertex and goal set	9
$\text{Exp}(A, G)$	Number of vertices algorithm A expands on graph G	9
$h(u, v)$	True minimum path cost from u to v , distinct from the heuristic estimate $\hat{h}(v)$	9
$\mathcal{I}_o$	Initiation set of option o : the states where the option may start	10
π_o	Option o 's internal policy, mapping execution information to a primitive action	10
β_o	Option o 's termination condition	10
μ	Policy over options: the parent policy that selects which option to run	10
a_t	Action selected at step t ; in Chapter 11, the next action selected after t completed pulls	11
$\text{Reg}(T)$	Regret accumulated over T rounds	11
$\hat{\mu}_a$	Empirical mean payoff of action a	11
$N_t(a)$	Number of pulls of action a among the t completed pulls, before the next decision	11
$\text{Gain}(O)$	Decision-relevant information in an observation	11
$\hat{V}_t(x)$	The agent's current estimate of the value of state x	12
δ_t	Temporal-difference error at step t	12
α	Step size for an update	12
λ	Weighting between the one-step estimate and the realized outcome	12
Ret_t^λ	The weighted return that interpolates between them	12
$\text{Pot}(x)$	Bounded shaping potential on declared state	13
r'_t, G'_0	Shaped reward and return; the original reward r and the return keep their meaning (Chapter 13 gives the return a local alias for Chapter 7's Ret_t)	13
ϕ	Trainable policy parameters, distinct from the frozen model parameters θ	13
$\hat{P}(x' x, a)$	Learned transition-model estimate	14

Symbol	Meaning	Introduced
$\widehat{\text{Obs}}$	Learned observation-kernel estimate	14
ϵ	Uniform one-step total-variation model error	14
τ_{samp}	Sampling temperature of a generative model	14
R	Upper bound on the common expected immediate state-action reward	14
H_{max}	Declared finite planning cap	14
R^*	Retained record set chosen under the token budget	15
$c(m), B, \lambda$ (Chapter 15)	Raw token cost of record m , the token budget, and decision-value units per token	15
$\epsilon, 2\epsilon$ (Chapter 15)	Summary value tolerance and the resulting regret bound	15
$\text{Cov}(k)$	Coverage at k : the chance that at least one of k samples is correct	16
$\text{Sel}(k)$	Selection at k : the chance that the returned sample is correct	16
$\hat{c}_{\text{success}}$	Total attempt cost divided by authorized confirmed completions	16
$\pi(k)$	Measured chance that the selector's top-ranked sample is correct given k candidates that contain at least one correct candidate	16
d	Probability that the selector scores one correct sample above one incorrect sample	16
$\text{eff}(T, x)$	State resulting from applying tool T in state x	17
$\text{resp}(T, x)$	Report returned by tool T in state x	17
$\text{undo}(T)$	Recovery tool for an effect of T	17
$\text{Rep}(x)$	Conservative set of eligible repeat attempts	17
$\text{Ready}(T, x)$	Current authorization and preconditions for a repeat attempt	17
$\text{Absent}(T, x)$	Authoritative terminal non-application of the original attempt	17
$\text{Restored}(T, x)$	Verified recovery of relevant consequences and preconditions	17
$\beta, c_{\text{dup}}, c_{\text{miss}}$	Belief that the effect already landed, cost of a duplicate effect, and cost of the effect never happening	17
$\text{Exec}_{\text{ui}}(\tilde{a} \mid x, a)$	Normalized realized-operation law	18
P_{ui}	Execution-composed world kernel	18

Symbol	Meaning	Introduced
Act_{ui}	Finite realized-operation set including declared denial/no-effect operations	18
$\mathcal{A}_{\text{rob}}(\mathbf{b})$	Intersection of correctly specified authorization sets over belief support	18
$\lambda_{\text{ui}}, \text{Fresh}(\Delta)$	Material-change rate and no-invalidating-change probability	18
$\mathcal{A}_{\text{auth}}(x)$	Commands whose protected effect is authorized in state x	18
$u_i(\pi_1, \pi_2)$	Payoff to party i under a pair of policies	19
$\pi_i^{(s)}$	Party i policy from training run s	19
$\text{diag}(M), \text{off}(M)$	Matched-run and cross-run mean returns	19
$\text{JPC}(M)$	Proportional cross-play loss	19
$\text{BR}_i(\pi_{-i})$	Best response by party i to the other policy	19
M	Matched matrix whose entry $M_{\{st\}}$ pairs run s against run t	19
$\text{Know}_i(F)$	Party i knows proposition F	20
$\text{MK}_{\mathcal{P}}^n(F)$	n levels of mutual knowledge among party set $\mathcal{P}$	20
$\text{CK}_{\mathcal{P}}(F)$	Common knowledge of F among party set $\mathcal{P}$	20
q_e	Nonnegative flow on traffic-network edge e	21
$\ell_e(q_e)$	Latency incurred on edge e at flow q_e	21
$\text{TL}(q)$	Total latency of a feasible network flow q	21
q^{NE}	Nash-equilibrium flow	21
q^*	Total-latency-minimizing flow	21
$\ell_e^{\text{mc}}(q_e)$	Marginal-cost latency imposed by edge e	21
$q_{\text{upper}}, q_{\text{lower}}, q_{\text{middle}}$	Flows on the three directed Braess routes S-U-T, S-L-T, S-U-L-T	21
γ_{cap}	Extra-traffic fraction in the bicriteria routing comparison	21
C	Declared episode-cost random variable for tail-risk accounting	22
$\text{CVaR}_{\alpha}(C)$	Conditional value at risk of C at declared tail level α	22
z	Candidate tail cutoff in CVaR's infimum representation	22
B, b_t	Declared bounded-horizon risk budget and remaining budget before decision t	22

Symbol	Meaning	Introduced
$\hat{r}_t(a_t)$	Conservative charged risk estimate for action a_t at time t	22
Ret^π	Expected discounted reward under policy π	22
$\mathcal{I}$	Set of states in which a named invariant holds	22
Cost_i^π	Expected discounted auxiliary-cost return	22
d_i	Declared limit on auxiliary cost i	22
$\mathcal{T}$	Declared threat model	22
ϵ_{safe}	Deliberate safety margin	22
$A_{C_i}^{\pi_k}(s, a)$	Constraint advantage for auxiliary cost i	22
ϵ_i	Largest candidate-policy expected constraint advantage	22
δ	Ideal CPO step-size bound	22
λ_{safe}	Fixed multiplier between reward and one auxiliary cost	22
$\mathcal{C}_{\text{parent}}, \mathcal{C}_{\text{child}}$	Declared capability sets of a delegating parent and a delegated child	23
M_t, e_t	Monitor's enforcement-layer record at time t , and one observed event	23
Mon	Monitor update map from prior record and one observed event to the next record	23
δ, v, r	Document identity, version, and recipient in the publish predicate	23
$\text{Approved}(M_t)$	Set of (δ, v, r) triples the monitor's current record authorizes for release	23
$\text{Publish}(\delta, v, r, M_t)$	Boolean predicate: is this exact document-version-recipient triple currently approved	23
$\mathcal{F}$	One declared, finite attack family under a fixed threat model	23
$\text{Risk}(\mathcal{F})$	Worst-case violation probability across $\mathcal{F}$	23
k	Number of samples drawn	24
p	Probability that one sample is correct	24
N_{eval}	Number of evaluated task instances	24
$\Delta_{\text{match}}(\theta)$	Score difference for frozen model θ between a named benchmark and its matched evaluation	24

Symbol	Meaning	Introduced
ζ	Declared threshold on a continuous metric when defining a thresholded scale onset	24
Onset_ζ	First declared family scale at which the thresholded metric reaches ζ	24
$\mathcal{K}$	Fixed same-bank task set used to compare coverage with deployed selection	24
$\widehat{\text{Cov}}_\mathcal{K}(k), \widehat{\text{Sel}}_\mathcal{K}(k)$	Empirical same-bank coverage and deployed-selector success after k candidates per task	24
$\mathcal{C}, \xi$	Finite tested configuration set and one configuration index	24
Pr	Probability over Equation 24.4's declared sampling or resampling design	24
$\text{Lower}_{\alpha_{\text{tail}}}^{\text{sim}}(\xi)$	Lower score bound for configuration ξ from a procedure with joint coverage across all tested configurations	24
SE_{match}	Estimated standard error of a difference of independent bank pass proportions	24
α_{tail}	Tail probability level for the simultaneous lower-bound guarantee	24
$\text{LCF}_{\alpha_{\text{tail}}}$	Simultaneous lower confidence frontier across tested configurations	24
ϕ_t, ϕ'	Parent and candidate versions of the agent procedure around frozen weights	25
$\widehat{\Delta}_G(\phi'), \mathbf{G}$	Guard-set estimate and its frozen guard cases for the named candidate	25
$\widehat{C}_G(\phi'), \mathcal{G}$	Guard cost estimate and authority envelope for the proposed release	25
$\tau, \mathbf{C}, \varepsilon_{\text{safe}}$	Local uplift threshold, cost limit, and reserved safety margin	25
$\mathbf{D}, \widehat{V}_D$	Development set and a procedure's estimated value on it	25
$\mathcal{M}, \Phi_t$	Declared bounded family of procedure changes and the finite candidate collection compared on $\mathbf{D}$	25
$\mathbf{m}, Z_i(\phi')$	Number of guard cases and the paired candidate-minus-parent outcome on guard case i , bounded in $[-1, 1]$	25
$\Delta(\phi')$	Population mean of $Z_i(\phi')$: the candidate's true uplift, an unknown parameter	25
$\text{Authorized}_\mathcal{G}, \text{Rollback}$	Boolean authority predicate for the proposed release and the check for a tested path back to the parent	25
$\mathbf{n}, \mathbf{c}$	Single-task aliases of $\mathbf{n}_i, \mathbf{c}_i$, the source-pool size and the number of passing samples for task i	26

Symbol	Meaning	Introduced
$\text{Del}(d, \Delta)$	Delegation action to destination d , with stated maximum delay Δ before a response, expiration, or fallback	27
$p_E(x)$	Probability that delegated destination E returns a correct resolution at input or state x	27
$\mathcal{R}_{\text{auth}}(x)$	Authorized decision routes in state x	27
ϱ	One member of $\mathcal{R}_{\text{auth}}(x)$, an authorized decision route	27
$V(\varrho, x)$	Declared value of authorized route ϱ in state x	27

Chapter 25 local notation

Chapter 25 uses $\phi_t, \phi', G, \hat{\Delta}_G, \hat{C}_G, \tau, c, \varepsilon_{\text{safe}}$, and $\mathcal{G}$ only for its constructed procedure-release rule. θ keeps its book-wide meaning: frozen model parameters. These local release symbols do not rename the book's standing state, action, score, or authority notation.

Notes on symbols in Chapters 13, 16, and 18

$\text{Pot}(x)$ in Chapter 13 is a bounded shaping potential; its terminal boundary affects policy comparisons. Chapter 16's cost per authorized confirmed success is undefined when no run succeeds. In Chapter 18, Exec_{ui} connects a proposed command to a realized operation. $\mathcal{A}_{\text{rob}}$ retains actions feasible throughout the declared belief support. $\text{Fresh}(\Delta)$ describes no material interface change under a stated constant-rate model.

Using the guide

Local indices, declared aliases, and diagram labels retain their stated scope; return to the defining chapter for domains and assumptions.